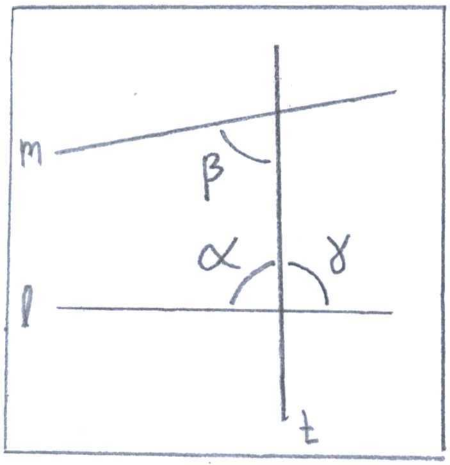
Chapter 4: Neutral Geometry

Exercises:

1\. a) (pg 173) Euclid's Postulate V



"Euclid's Postulate V"

"If two lines are intersected by a traversal in such a way that the sum of the degree measures of the two interior angles on one side is less than 180°, then the two lines meet on that side of the traversal."

Converse to Euclid's V

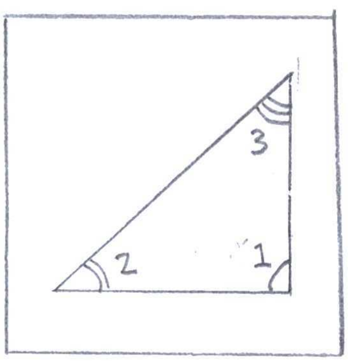
"If two lines are intersected by a traversal in such a way that they meet, then the sum of the degree measures of the two interior angles on one side is less than 180°.

Proof:

Hypothesis	Evidence
1) $\measuredangle\alpha + \measuredangle\beta < 180°$	RAA hypothesis
2) $\measuredangle\alpha + \measuredangle\gamma = 180°$	Theorem 4.3(5) (supplementary angles)
3) $\measuredangle\beta < 180° - \measuredangle\alpha$ $= \measuredangle\gamma$	Substitution

4) A horizontal ray $\overrightarrow{m}$ exists at $\measuredangle\beta$	Axiom C-4
5) $\overrightarrow{m} \parallel \overrightarrow{l}$	Euclid Postulate V
6) Lines $\overrightarrow{l}$ and $\overrightarrow{m}$ meet at the opposite side, since $\measuredangle\beta > \measuredangle\gamma$	Theorem 4.2
7) Lines $\overrightarrow{l}$ and $\overrightarrow{m}$ meet on the same side as $\measuredangle\alpha$ and $\measuredangle\beta$	RAA result by contradiction.

b) (pg 165) Corollary 1 to Exterior Angles
"If a triangle has a right or obtuse angle, the other two angles are acute"

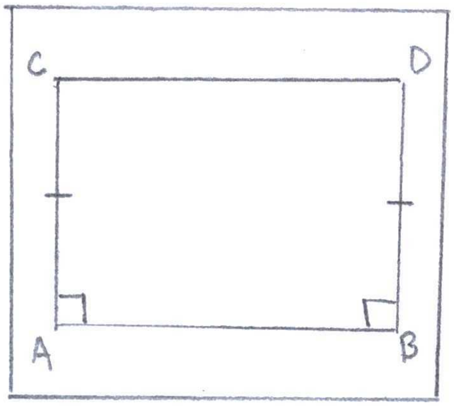


"right or obtuse"

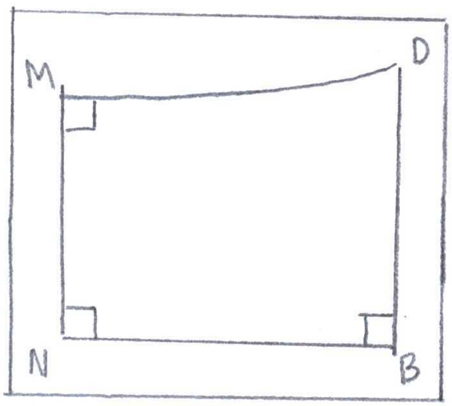
Proof: Hypothesis	Evidence
1) An angle ($\measuredangle 1$) is right or obtuse in a triangle	Measurement Theorem 4.3(1)
2) $\measuredangle 2 + \measuredangle 3 \leq 90°$	Proposition 4.11
3) $\measuredangle 2$ and $\measuredangle 3$ are real in addition to acute	Measurement Theorem 4.3(4)

c) (from exercise) "All Saccheri and Lambert quadrilaterals are rectangles and.. rectangles exist.

Proof



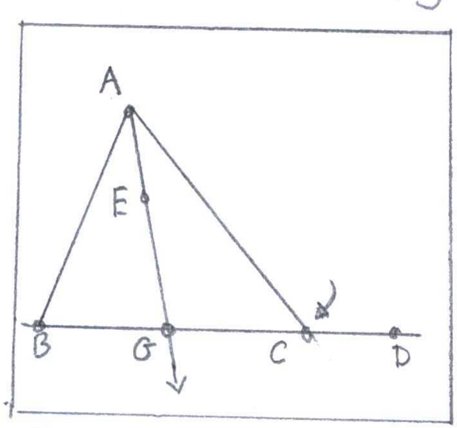
"Saccheri Quadrilateral"



"Lambert Quadrilateral"

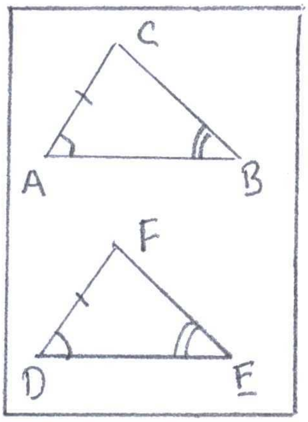
Hypothesis	Evidence
1) All Saccheri quadrilateral □ABCD is a Lambert quadrilateral □NBMD	Saccheri's Angle Theorem (His Proposition $\overline{XV}$)(c).
2) A perpendicular from C to $\overleftrightarrow{BD}$ is point D	Proposition 4.4 (Bisectors)
3) Otherwise another parallel line exists to $\overleftrightarrow{AB}$ through C	Corollary 1 to A.I.A. theorem

d) (pg 186) "In a Hilbert Plane satisfying Aristotles axiom, a exterior angle of a triangle is greater than or congruent to the sum of the two remote interior angles."



"exterior angle"

2. (from exercise) "Proof in Neutral Geometry



"SAA vs. ASA"

Given $AC \cong DF$, $\measuredangle A \cong \measuredangle D$, $\measuredangle B \cong \measuredangle E$,

Then $\measuredangle C \cong \measuredangle F$ since

$$(\measuredangle C)^\circ = 180^\circ - (\measuredangle A)^\circ - (\measuredangle B)^\circ$$
$$= 180^\circ - (\measuredangle D)^\circ - (\measuredangle E)^\circ = (\measuredangle F)^\circ$$

Hence $\triangle ABC \cong \triangle DEF$ by ASA.

*The flaw in Neutral geometry is ASA Versus SAA.

**Also, in Elliptic or Hyperbolic geometry, angles in a triangle are not 180-degrees.

3. (pg 166) Proposition 4.1 (SAA Criterion)

"Given $AC \cong DF$, $\measuredangle A \cong \measuredangle D$, and $\measuredangle B \cong \measuredangle E$

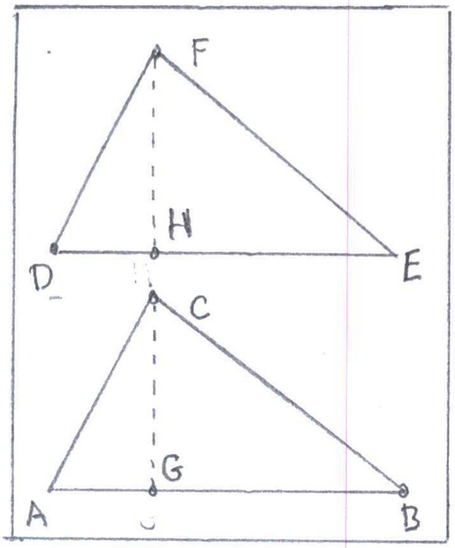
Then $\triangle ABC \cong \triangle DEF$"

Proof

Hypothesis	Evidence
1) Side AB is not congruent to side DE	1) RAA False Conjecture

Proof:

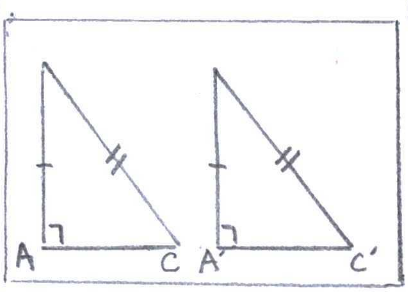
Hypothesis	Evidence
Case 1: $\sphericalangle BAC \cong \sphericalangle ACD$	
1) $\overleftrightarrow{AB}$ is parallel to $\overleftrightarrow{CD}$	Theorem 4.1 contradiction
Case 2: $\sphericalangle BAC > \sphericalangle ACD$	
1) Ray $\overrightarrow{AE}$ is between $\overrightarrow{AB}$ and $\overrightarrow{AC}$, such $\sphericalangle ACD \cong \sphericalangle CAE$	RAA conjecture
2) Ray $\overrightarrow{AE}$ intersects BC at point G	Crossbar Theorem
3) Impossible because $180° = \sphericalangle A + \sphericalangle B + \sphericalangle C$ $= \sphericalangle BAC + \sphericalangle CBA + \sphericalangle ACB$ $= (\sphericalangle CAE + \sphericalangle EAB) + \sphericalangle CBA + \sphericalangle ACB$ $= (\sphericalangle ACD + \sphericalangle EAB) + \sphericalangle CBA + \sphericalangle ACB$ leads to 0°-angles	Contradiction to RAA conjecture. because supplementary angles
Case 3: $\sphericalangle BAC < \sphericalangle ACD$	
1) $\sphericalangle ACD \geq \sphericalangle BAC + \sphericalangle CBA$	Proposition 3.21(A)



"Proof of the SAA criterion"

2) Then AB < DE or DE < AB	Proposition 3.13 (Segment Ordering)
3) If DE < AB, then point G is between A and B such that AG ≅ DE	Euclid's Postulate II
4) Then △CAG ≅ △FDE	Proposition 3.17 (ASA criteria for congruence)
5) Hence ∢AGC ≅ ∢E and ∢AGC ≅ ∢B	RAA Result
6) Therefore, DE is not less than AB	Implication to a false conjecture
7) By similar argument a point H is between D and E.	Axiom B-2
8) Hence AB ≅ DE, and △ABC ≅ △DEF	Euclid Postulate II

4.



"Hypotenuse - leg"

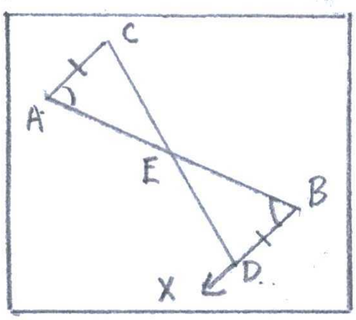
(pg 147) Proposition 4.2 (Hypotenuse-Leg Criterion)

"The right triangles are congruent if the hypotenuse and leg of one are congruent, respectively to the hypotenuse and leg of the other."

Proof:

Hypotenuse	Evidence
1) Three distinct points form a triangle: A, B, C.	(pg 90) "Definition - Triangle"
2) Three distinct points form another triangle: A', B', C'.	(pg 90) "Definition - Triangle"
3) A ray emanates from AC through point D	(pg 18) "Definition - Ray"
4) $AB = A'B'$, $BC = B'C'$ and $\measuredangle A = \measuredangle A'$, so $\Delta DAB = \Delta A'B'C'$	Axiom C-6 (S.A.S.)
5) But point D is C, $\Delta ABC = \Delta A'B'C'$	False conjecture in Step #4 leads to a reality.

5.



"midpoints"

(pg 167) Proposition 4.3 (Midpoints)

"Every segment has a midpoint"

Proof

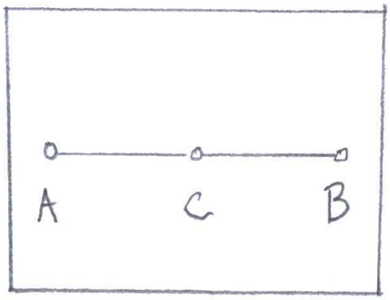
Hypothesis	Evidence
1) Let C be any	Axiom I-3

point not on $\overleftrightarrow{AB}$	
2) A unique ray $\overrightarrow{BX}$ on the opposite side of $\overleftrightarrow{AB}$ from C such that $\sphericalangle CAB \cong ABX$	Axiom C-4
3) There is a unique point D on $\overrightarrow{BX}$ such $AC \cong BD$.	Axiom C-1
4) D is on the opposite side of $\overleftrightarrow{AB}$ from C.	Axiom B-4
5) Let E be the point when segment CD intersects AB.	Proposition 2.1
6) Assume E is not between A and B on $\overleftrightarrow{AB}$	RAA #1 conjecture
7) Then either $E=A$, or $E=B$, or $E*A*B$, or $A*B*E$.	Axiom B-2

8) $\overleftrightarrow{AC}$ is parallel to $\overleftrightarrow{BD}$.	Euclid's Postulate V
9) Hence, $E \neq A$, and $E \neq B$	RAA #1 outcome
10) $E * A * B$ assuming	RAA #2 conjecture
11) Since $\overleftrightarrow{CA}$ intersects side EB or BD.	RAA #2 outcome
12) Yet, the statement above was impossible. Hence A is not between E and B.	True outcome
13) Similarly, B is not between A and E. Thus, $A * E * B$. Also, $\angle AEC \cong \angle BED$.	True Outcome
14) $\triangle EAC \cong \triangle EBD$ along with E the midpoint of AB.	Proposition 4.1 (S.A.A. conjecture criterion).

6. a) (from exercise) "Segment $\overline{AB}$ has only one midpoint."

Proof:



"midpoints"

Hypothesis	Evidence
1) Assumption, $\overline{AB}$ has no midpoint.	RAA conjecture, but false.
2) On segment $\overline{AB}$ is a distinct point C between A and B.	Axiom B-1
3) Points A, B, C form $A * C * B$ on segment AB.	Proposition 3.3
4) $\overline{AC} < \overline{CB}$ or $\overline{AC} > \overline{CB}$	Proposition 3.13
5) Previous statement is false, so $\overline{AB}$ has a midpoint.	RAA result after argument.

b) (pg 168) Proposition 4.4 (Bisectors)

A) Every angle has a unique bisector

B) Every segment has a unique perpendicular bisector.

Proof:

Hypothesis	Evidence
1) Three unique points reside in space: A, B, C.	(pg 12) "Definition - point"
2) The three points form an angle, $\measuredangle ABC$	(pg 18) "Definition - angle"
3) $\overline{AB} = \overline{BC}$	(pg 11) "Requirement 0"
4) Rays $\vec{x}$ and $\vec{y}$ are bisectors.	RAA hypothesis.
5) $\vec{x}$ and $\vec{y}$ intersect $\overline{AC}$ at point P and Q, so $\measuredangle ABP \cong \measuredangle CBP$, and $\measuredangle ABQ \cong \measuredangle CBQ$	Proposition 4.4 (Bisectors)
6) $\overline{BP} = \overline{BQ}$	Euclid Postulate II
7) $\Delta BAX \cong \Delta CAX$ and $\Delta BAY \cong \Delta CAY$	Axiom C-6 (S.A.S.)
8) $\overline{AP} = \overline{PC}$ and $\overline{AQ} = \overline{QC}$	(pg 119) "Definition - congruent"

9) But $P=Q$ with a midpoint to $\overline{AC}$.	RAA contradiction.

17. (from exercise) "Every acute angle has a complementary angle."

Proof:

Hypothesis	Evidence
1) A right angle forms from points A, B, C to $\measuredangle ABC$.	(pg 11) "Definition - Right angle"
2) A point D is in the interior of $\measuredangle ABC$.	Proposition 3.7
3) D makes an acute angle $\measuredangle DBC$	(pg 129) "Definition - acute angle"
4) $\measuredangle DBA$ is complementary to $\measuredangle DBC$	(pg 171) "Definition - Complementary angle"

(from exercise) "If compliments of two acute angles are congruent, then the acute angles are congruent."

Proof

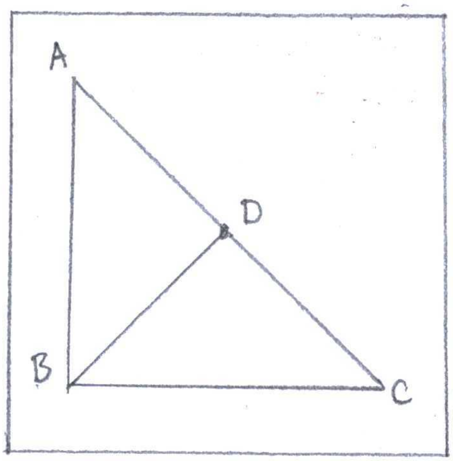
Hypothesis	Evidence

Statement	Reason
1) A right angle forms ∡ABC	(pg 11) "Definition - right angle"
2) A point D is in the interior of ∡ABC	Proposition 3.7
3) D makes an acute angle ∡DBC	(pg 129) "Definition - acute angle"
4) ∡DBA is complementary to ∡DBC	(pg 171) "Definition - complimentary angle"
5) Another right angle ∡EFG has an interior point H, where ∡HFG acute and ∡HFG compliment to ∡HFE.	(pg 11, 129, 171) "Definitions - right angle, interior, acute angle, complimentary angle"
6) ∡DBA ≅ ∡HFE	RAA hypothesis
7) ∡DBC ≅ ∡HFG	Proposition 3.20 (Angle Subtraction)

8. (pg 168) Proposition 4.5

In a triangle △ABC, the greater angle lies opposite the greater side and the greater side lies opposite the greater angle; i.e, $AB > BC$ if and only if $\measuredangle C > \measuredangle A$."

Proof

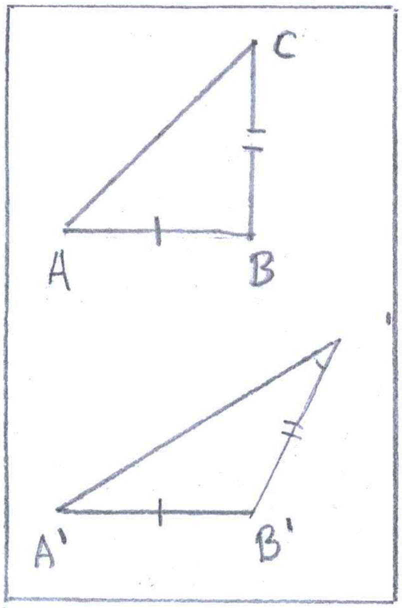


"greater side... opposite greater angle"

Hypothesis	Evidence
Greater Angles:	
1) Triangle ΔABC appears with point D on side AC where $AB = AD$	(pg 98) "Definition-Triangle"
2) Angle $\measuredangle ABC$ is greater than $\measuredangle BCA$.	RAA conjecture
3) $\measuredangle ADB > \measuredangle DCB$	(pg) "Definition-Exterior Angle"
4) $\measuredangle ABD = \measuredangle ADB$	Proposition 3.X (isoscles)
5) $\measuredangle ABD > \measuredangle DCB$	Measurement Theorem (A.6)
6) $\measuredangle ABC = \measuredangle ABD + \measuredangle BC$	Proposition 3.X (Angle Addition)
7) $\measuredangle ABC > (\measuredangle BCA = \measuredangle DCB)$	RAA outcome
Greater Sides:	
Case 1: $AC = AB$	RAA conjecture #1
A) $\measuredangle ABC = \measuredangle ACB$	RAA false outcome #1 $(T \to F)$
Case 2: $AC < AB$	RAA conjecture #2
A) $\measuredangle ABC < \measuredangle ACB$	RAA false outcome #2

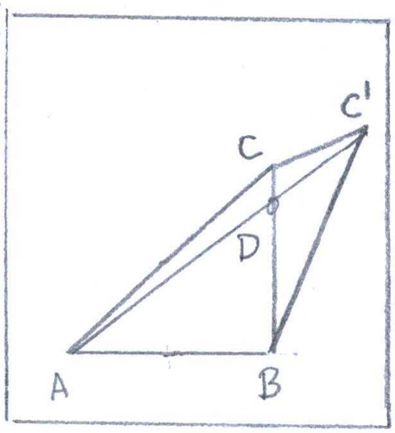
Hypothesis	Evidence
Case 3: $AC > AB$	RAA conjecture #3
A) $\angle ABC > \angle ACB$	RAA conclusion

9.



"greater side... opposite angle"

necessary figures.



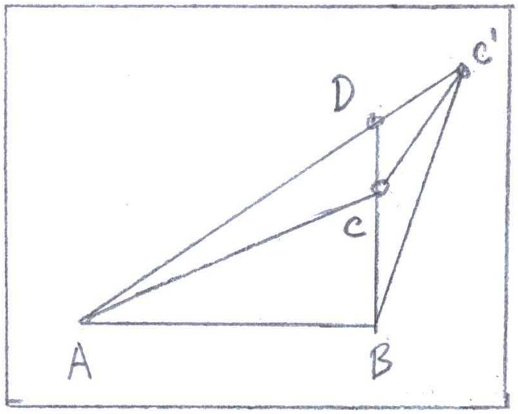
"... case 1"

(pg 168) Proposition 4.6

"Given $\triangle ABC$ and $\triangle A'B'C'$ if we have $AB \cong A'B'$ and $BC \cong B'C'$, then $\angle B < \angle B'$ if and only if $AC < A'C'$.

Proof:

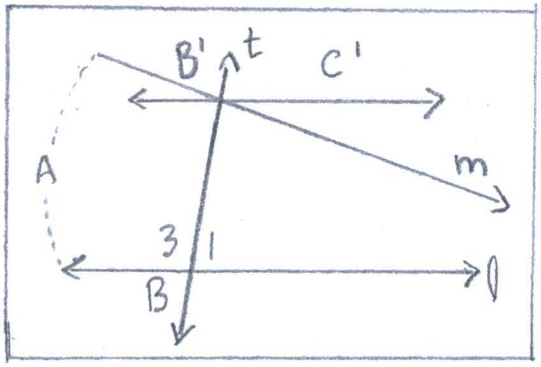
Hypothesis	Evidence
1) In an instance, points $A = A'$, $B = B'$, and angle $\angle B < \angle B'$	RAA hypothesis
2) Point C is the interior to $\angle ABC'$	Proposition 3.7
Case 1: $C = D$	
A) Point C is not unique	(pg 19) "Definition - Point"
B) $\angle ABC < \angle ABC'$	Proposition 4.5
C) $AC < AC'$	RAA outcome
Case 2: $C \neq D$	
A) $BC = BC'$	Axiom C-1
B) $\angle AC'C < \angle ACC'$	Proposition 4.5
Sub-case: $B*D*C$	



"... case 2"

I) Points B, D, and C form a line	Axiom B-1
II) $\measuredangle BCC' < \measuredangle ACC'$	RAA result
Sub-case: $B * C * D$	
I) Points B, C, and D form a line, $B * C * D$	Axiom B-1
II) $\measuredangle ACC' > \measuredangle DCC'$	Measurement Theorem (A.6
III) $\measuredangle ACC' > \measuredangle BC'C$	Measurement Theorem (A.6)
IV) $\measuredangle BC'C = \measuredangle BCC'$	Measurement Theorem (A.2)
V) $\measuredangle ACC' > \measuredangle BCC'$	Measurement Theorem (A.6)
VI) $\measuredangle BCC' > \measuredangle CC'D$	Measurement Theorem (A.6)
VII) $\measuredangle CC'D = \measuredangle AC'C$	Measurement Theorem (A.2)
VIII) $\measuredangle ACC' > \measuredangle AC'C$	Measurement Theorem (A.6)
3) All cases show $AC < AC'$	Exterior Angle Theorem (4.2)

10.



"Hilbert's Euclidean Parallel Postulate"

(pg 175) Proposition 4.7

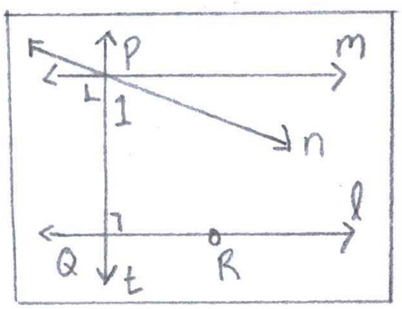
"Hilberts Euclidean parallel postulate $\Leftrightarrow$ if a line intersects one of two parallel lines, then it also intersects the other"

Proof: Hypothesis	Evidence
1) Angles $(\measuredangle 1)^\circ$ and $(\measuredangle 2)^\circ$ sum to 180°	RAA conjecture
2) $(\measuredangle 1)^\circ + (\measuredangle 3)^\circ = 180^\circ$	Measurement Theorem 4.3(5)
3) $(\measuredangle 2)^\circ < (180^\circ - (\measuredangle 1)^\circ) = (\measuredangle 3)^\circ$	RAA conjecture chained
4) A ray $\overrightarrow{BC}$ passes $(\measuredangle 3)^\circ$ and $\measuredangle C'B'B$ in a way to congruent alternate angles.	Axiom C-4
5) $\overleftrightarrow{B'C'}$ is parallel to $\overleftrightarrow{l}$	Alternate Interior Angles (AIA) Theorem 4.1
6) Line $\overleftrightarrow{m}$ is not $\overleftrightarrow{B'C'}$, so m meets	Corollary to Euclid's Postulate V

$\overleftrightarrow{l}$ when $(\measuredangle 2)^\circ < (\measuredangle 3)^\circ$

7) m meets $\overleftrightarrow{l}$ on the same side of line $\overleftrightarrow{t}$ or $(\measuredangle 2)^\circ$ an exterior angle of $\Delta ABB'$	False RAA outcome in a compound statement by chaining. $(T \to T \to F)$
8) Whither, $(\measuredangle 2)^\circ < (\measuredangle 3)^\circ$ and not an exterior angle	True RAA outcome by contradiction Exterior Angle (EA) Theorem 4.2.

11.



"Hilbert's... Parallel Postulate"

(pg 175) Proposition 4.8

"Hilbert's Euclidean parallel postulate $\Leftrightarrow$ converse to alternate interior angle theorem"

Proof:

Hypothesis	Evidence
1) Lines $\overleftrightarrow{m}, \overleftrightarrow{n}$, and $\overleftrightarrow{l}$ are parallel	Euclid's Postulate V
2) $\overleftrightarrow{m}$ and $\overleftrightarrow{n}$ pass through point P	Proposition 2.1
3) Line $\overleftrightarrow{m}$ is certainly parallel to $\overleftrightarrow{l}$	(pg 163) Corollary 2 to Alternate Interior Angle Theorem

4) The alternate interior angles are congruent by traversal $\overleftrightarrow{n}$	(Euclid I. 31) Converse to Alternate Interior Angle Theorem

12. (pg 175) Proposition 4.9.

"Hilberts Euclidean parallel postulate $\Longleftrightarrow$ if t is a traversal to l"

Proof:

Hypothesis	Evidence
1) Line $\overleftrightarrow{t}$ is perpendicular to $\overleftrightarrow{l}$ through Q and line m through P	RAA hypothesis
2) $\overleftrightarrow{m} \parallel \overleftrightarrow{l}$	Corollary 1 to Theorem 4.1
3) $\overleftrightarrow{n}$ traverses point R, but not $\overleftrightarrow{l}$	Proposition 2.1
4) $\measuredangle PQR$ is $90°$	(pg 19) "Definition - Right angle"
5) $(\measuredangle 1)° + (\measuredangle PQR)° = 180°$	Measurement Theorem 4.3 (A.5)
6) $\measuredangle 1 = 90°$ and $\overleftrightarrow{t} \perp \overleftrightarrow{m}$	RAA outcome

13. (pg 175) Proposition 4.10

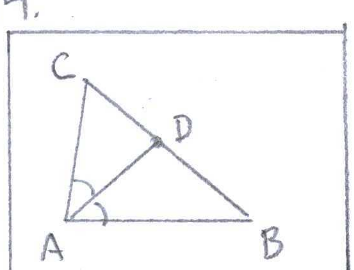
"Hilberts Euclidean parallel postulate $\Leftrightarrow$ if $k \parallel l$, $m \perp k$, and $n \perp l$, then either $m = n$ or $m \parallel n$."

Proof:

Hypothesis	Evidence
1) Line $\overleftrightarrow{k}$ is parallel to $\overleftrightarrow{l}$	Euclids Postulate V
2) Line $\overleftrightarrow{m}$ and $\overleftrightarrow{l}$ are perpendicular to $\overleftrightarrow{k}$.	(pg 19) "Definition - right angle"
3) Line $\overleftrightarrow{n}$ is perpendicular to $\overleftrightarrow{l}$	(pg 19) "Definition - right angle"
4) $\overleftrightarrow{m} \perp \overleftrightarrow{l}$ and $\overleftrightarrow{n} \perp \overleftrightarrow{k}$	Proposition 4.9
5) $\overleftrightarrow{m} \parallel \overleftrightarrow{n}$ or $\overleftrightarrow{m} = \overleftrightarrow{n}$	Corollary 1 to Proposition 4.1

14. (from exercise) Heron's Formula

$$\overline{AB} + \overline{AC} > \overline{BC}$$



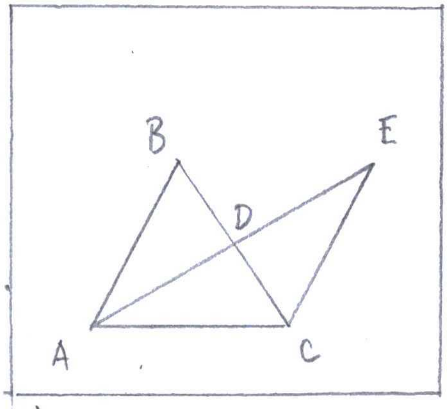
"Heron's Formula"

Proof:

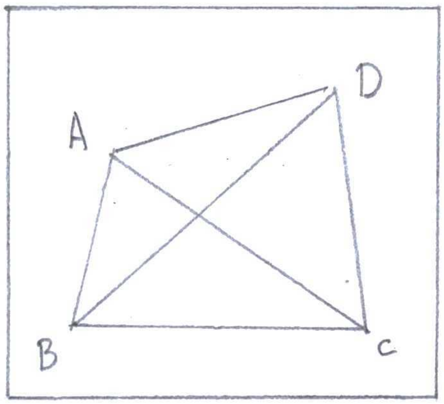
Hypothesis	Evidence
1) Points A, B, and C	

form a triangle.	(pg 90) "Definition - Triangle"
2) $\measuredangle A$ has a bisector	Proposition 4.4 (Bisector)
3) The bisector has a segment from point A to D.	(pg 109) "Definition - segment"
4) D is between AB and BC	Crossbar Theorem
5) $\measuredangle B$ and $\measuredangle DAB$ are less than $\measuredangle BDE$	Exterior Angle Theorem (EA) 4.2 Also, Proposition 4.5
6) $\overline{AB}$ is opposite to $\measuredangle BDE$	Proposition 4.5
7) $\overline{AB} > \overline{BD}$ $\overline{AC} > \overline{DC}$	Proposition 4.5
8) $\overline{BC} = \overline{BD} + \overline{DC}$	Axiom C-3
9) $AB + AC > BD + DC$ $= BC$	RAA result

15. (pg 186) Saccheri-Legendre Theorem
"In an Archimedean Hilbert plane, the angle sum of every triangle is $\leq 180°$"



"Legendre's Lemma"



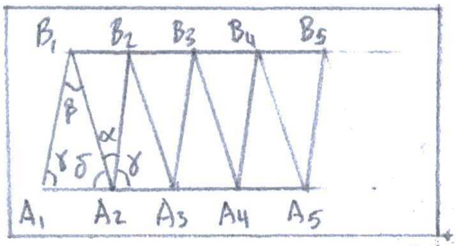
"Saccheri-Legendre Theorem"

Proof: Hypothesis	Evidence
1) ◻ABCD is a convex quadrilateral	(pg 199) "Definition- convex Quadrilateral"
2) $\measuredangle DCB = \measuredangle DCA + \measuredangle ACB$	(pg 19) "Definition- Supplementary Angles"
3) $\measuredangle DCA$ or $\measuredangle ACB \leq \frac{1}{2}(\measuredangle DCB)$	RAA hypothesis
4) The triangle ΔABC has angles greater than or equal to 180°	RAA result by false assumption $T \to F$ implied $\neg F$ and "less than".
5) The angle $\measuredangle BAC$ is no more than half $\measuredangle A$	Legendres Lemma
6) Three triangles are possible by another diagnol segment from $\measuredangle B$ to AC toward point D.	(pg 109) "Definition- Segment"
7) $\measuredangle BDC \leq \frac{1}{2} \measuredangle B$	Legendres Lemma
8) By consecutive diagnol segments, each angle is greater than 180°	RAA Conclusion $T \to F \to F$ implies false argument and "less than."

16.

(from exercise) Saccheri-Legendre Theorem

"Angle sum of every triangle is $\leq 180°$ in an Archimedean Hilbert plane"

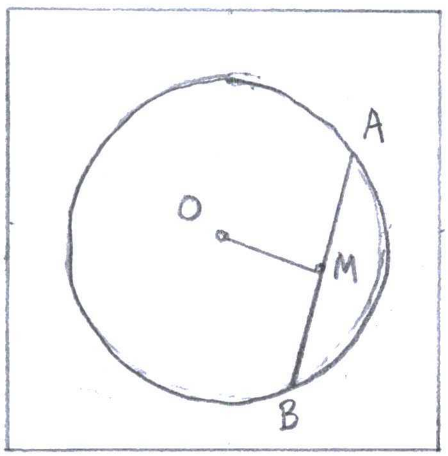


"Sacheri-Legendre Theorem" Proof:

Hypothesis	Evidence
1) Points $A_1A_2B_1$ are the triangle. Then lay off n copies of segment A_1A_2 and construct a row of triangles $A_jA_{j+1}B_j$, $j=1,\ldots,n$	RAA conjecture Archimedes Theorem
2) The $B_jA_{j+1}B_{j+1}$, $j=1,\ldots,n$, are also congruent triangles, the last construction of B_{n+1}	Axiom C-6 (SAS)
3) With angles in the figure, $\alpha+\gamma+\delta=180°$ and we have β to $\gamma+\delta$ equal to the angle sum $A_1A_2B_1$	Measurement Theorem 4.3(A.5

4) By the contrary, $\beta > \alpha$	RAA hypothesis
5) Then $A_1A_2 > B_1B_2$	Proposition 4.6
6) Also, $A_1B_1 + nB_1B_2 + \cdots B_{n+1}A_{n+1} > n \cdot A_1A_2$	Archimedes Axiom
7) $A_1B_1 \cong B_{n+1}A_{n+1}$	Axiom C-6
8) $2A_1B_1 > n(A_1A_2 - B_1B_2)$	Archimedes Axiom
9) n was arbitrary	False RAA outcome
10) Hence, the triangle has angle sum $\leq 180°$	... from argument.

17. a) (from exercise) "If $O \neq M$, then OM is perpendicular to AB"



"Chord"

Proof: Hypothesis	Evidence
1) A circle (γ) derives by center O and point A or B	Euclids Postulate III
2) A and B form a segment AB.	(pg 109) "Definition-segment"
3) AB has a midpoint M	Proposition 4.3 (Midpoints)
4) Segment AB has a unique	Proposition 4.4(b) (Bisectors)

perpendicular bisector.

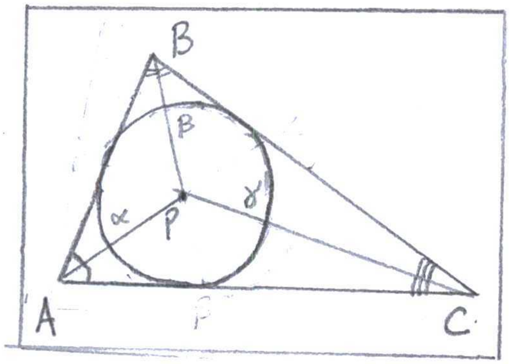
b) (from exercise) "A perpendicular bisector of AB passes through center O of γ"

Proof:

Hypothesis	Evidence
1) A circle(γ) derives from a center O and point A or B.	Euclids Postulate III
2) A and B form a segment $\overline{AB}$.	(pg 109) "Definition-segment"
3) AB has a mid-point M	Proposition 4.3 (Midpoint)
4) Segment $\overline{AB}$ has a unique perpendicular bisector through O & M	Proposition 4.4(b) (Bisectors)
5) O is a midpoint and γ center, especially when O=M!	Proposition 4.3 (Midpoint)

18. (from exercise) Incenter

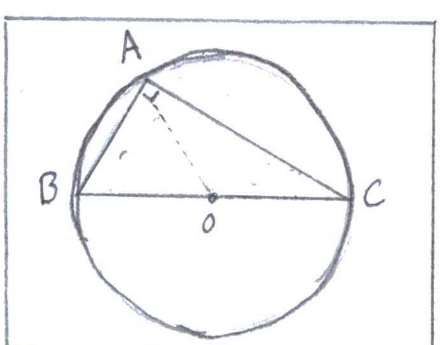
"Three angle bisectors are concurrent."



"Incenter"

Proof Hypothesis	Evidence
1) Three points form a triange; A, B, and C.	(pg 90) "Definition-Triangle"
2) AC and BC have unique bisectors α and β, accordingly.	Proposition 4.4(b)(Bisectors)
3) α and β inter-sect at a point P	Proposition 2.1
4) Point P also intersects a bi-sector from ∡C where $\alpha = \beta = \gamma$.	Measurement Theorem 4.3(B.3)

19.



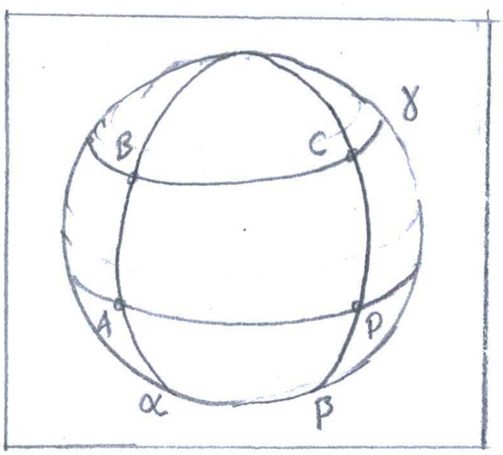
"Theorem of Thales"

(from exercise) "An angle inscribed in a semicircle is a right angle."

Proof: Hypothesis	Evidence
1) A circle at center O draws from point B or C	Euclids Postulate III

2) $B * O * C$	(pg 70) "Definition - collinear"
3) Segments $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ form a triangle	(pg) "Definition - triangle"
4) ΔOBA and ΔOAC are isoscles because $OA = OB = OC$	(pg) "Definition - Isoscles"
5) $\measuredangle OBA = \measuredangle OAB$ and $\measuredangle OAC = \measuredangle OCA$	Proposition 3.10
6) $\alpha + (\alpha + \beta) + \beta = 180°$, $\alpha + \beta = 90°$	I. 32

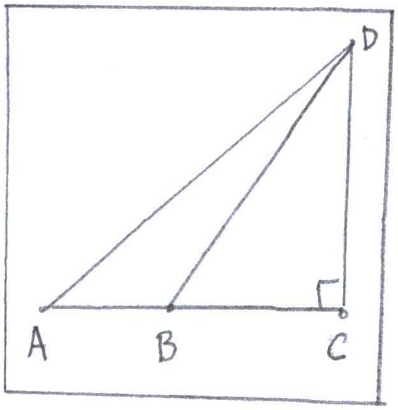
20.



"rectangle on a sphere"

The rectangle on a sphere is not standard in neutral geometry. Lemma from "Saccheri's Angle Theorem", "Let □ABCD be a saccheri quadrilateral with summit angle θ... The alternate interior angles $\measuredangle B$ and $\measuredangle D$ are not right angles." Neutral geometry requires a plane and total angle 360° in a rectangle.

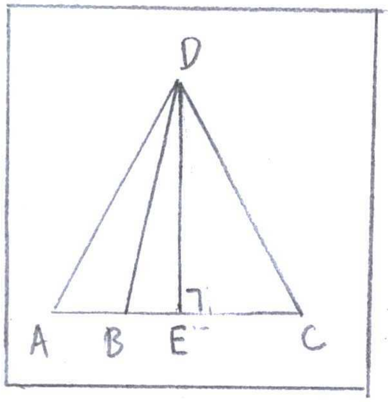
22.



(from exercise) "Given $A*B*C$ and $DC \perp AC$.
Prove $AD > BD > CD$."

Proof: Hypothesis	Evidence.
1) $BD > CD$ because $\measuredangle C > \measuredangle DBC$	Proposition 4.5
2) $\measuredangle C + \measuredangle DBC + \measuredangle CDB = 180°$	Proposition 4.11
3) $\measuredangle DBA = 180° - 90° - \measuredangle CDB \leq 90°$	(pg 19) "Definition - Supplementary angles."
4) $\measuredangle DBA > \measuredangle DBC$	
5) $AD > BD$ because $\measuredangle DBA > \measuredangle A$	Proposition 4.5
6) $AD > BD > CD$	Measurement Theorem (B.10)

23.



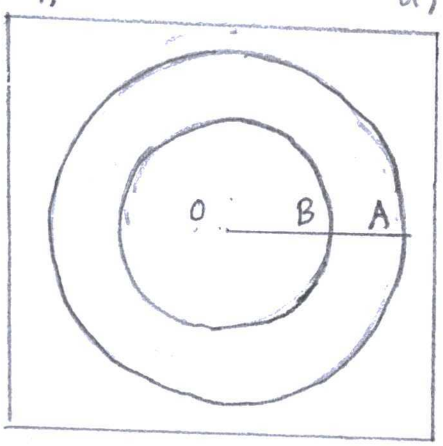
(from exercise) "$DB < DA$ or $DB > DA$"

Proof Hypothesis	Evidence
1) Three points make a triangle; A, B, C	(pg 98) "Definition - Triangle"
2) A point B appears on AC, so $A*B*C$	Axiom B-2.

3) Point E is between A and C, so $A * E * C$	Axiom B-2
4) A perpendicular bisector ED exists from AC.	Proposition 4.4(b)(Bisectors)
<u>Case 1: A and B are on the same side of ED:</u>	Axiom B-4
A) $BD > ED$ because $\angle DEB > \angle DBE$	Proposition 4.5
B) $\angle DEB + \angle DBE + \angle BDE = 180°$	Proposition 4.11
C) $\angle DBE = 180° - 90° - \angle BDE \leq 90°$	Measurement Theorem 4.3(A.1, A.6)
D) $DB < DC$ because $\angle DBE \leq (\angle DEC = 90°)$	Proposition 4.5
<u>Case 2: A and B are on opposite sides to ED:</u>	Axiom B-4

A) $\measuredangle DEB + \measuredangle DBE + \measuredangle BDE = 180^\circ$	Proposition 4.11
B) $\measuredangle DBE = 180^\circ - 90^\circ - \measuredangle BDE \leq 90$	Measurement Theorem 4.3(A.1, A.6)
c) $DB < DA$ because $\measuredangle DBE \leq \measuredangle DEA$	Proposition 4.5

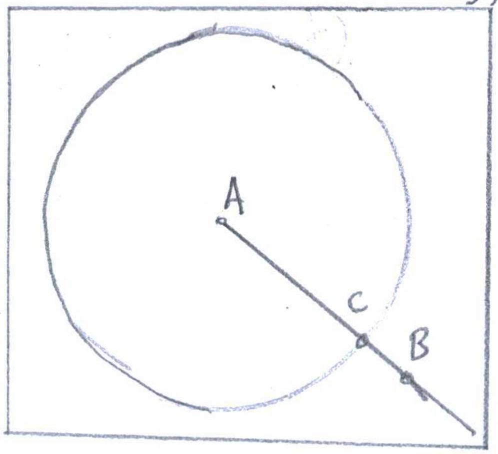
24.



a) (from exercise) "Interior of a circle is a convex set"

Proof:

Hypothesis	Evidence
1) A circle forms by center O and point A	Euclids Postulate III
2) Another point B is between O and A, so $O * B * A$	Axiom B-2
3) A second circle forms from OA	Euclids Postulate III
4) $OB \subseteq OA$	(pg)"Definition-subset"
5) Circle B $\subseteq$ Circle A	(pg)"Definition-subset"

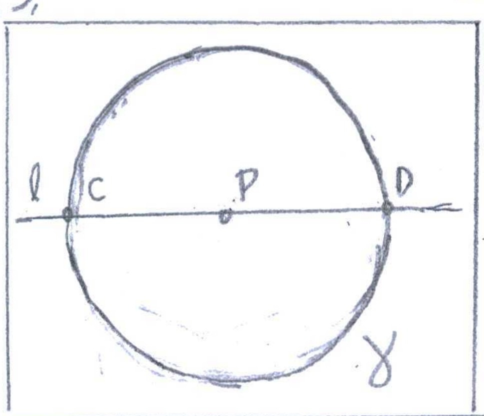


b) (from exercise) "A line passes through a point inside a circle, also it passes through points outside the circle."

Proof:

Hypothesis	Evidence
1) Two points A and B form a line	Euclid Postulate I
2) A point is between A and B, specifically C	Axiom B-2
3) Points A and C are the radius to a circle	Euclid Postulate III

25.

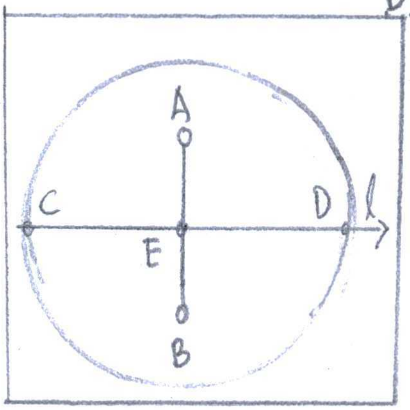


a) (from exercise) "Point P and ℓ lie inside γ if and only if $C * P * D$"

Proof:

Hypothesis	Evidence
1) Two points make a line (ℓ): C and D	Euclid Postulate I
2) The midpoint O between CD exists.	Proposition 4.3 (Midpoints)
3) A circle γ draws	Euclids Postulate III

3) A circle γ draws by OC or OD	Euclids Postulate III
4) A point P appears on l.	Axiom B-2.
<u>Case 1: $P*C*O$</u>	RAA hypothesis #1
A) $PO > CO$	Proposition 3.13 (Segment Ordering)
B) Point P is not in γ	RAA result #1 (False)
<u>Case 2: $C*P*O$</u>	RAA hypothesis #2
A) $PO < CO$	Proposition 3.13 (Segment Ordering)
B) Point P is in γ	RAA result #2 (True)
<u>Case 3: $O*P*D$</u>	RAA hypothesis #3
A) $OP < OD$	Proposition 3.13 (Segment ordering)
B) Point P is in γ	RAA result #3 (True)
<u>Case 4: $O*D*P$</u>	RAA hypothesis #4
A) $OP > OD$	Proposition 3.13 (segment ordering)
B) Point P is not in γ	RAA result #4 (False)



b) (from exercise) "If points A and B are inside γ and on opposite sides of l, then point E at which AB meets l is between C and D"

Proof

Hypothesis	Evidence
1) Two points make a line l; C and D	Euclids Postulate I
2) The midpoint between CD exists; O	Proposition 4.3 (midpoint)
3) A circle γ draws by OC or OD	Euclid Postulate III
4) Points A and B are inside γ	(pg 115) "Definition - interior"
5) A and B are opposite sides of l	Axiom B-4
6) $AO < CO$ and $OB < CO$, so $EO < CO$	Proposition 3.13 (segment ordering)

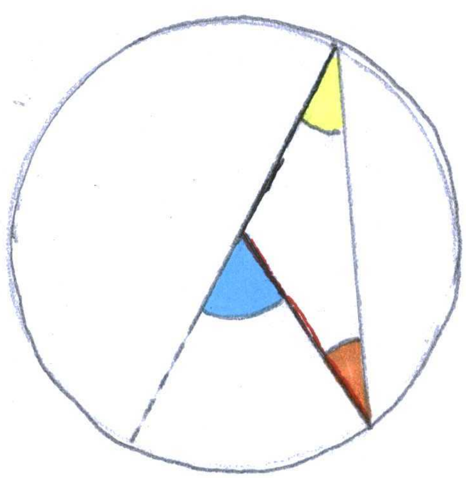
26. Cite: Byrne, O.

"The Elements of Euclid in which Coloured diagrams and symbols are used instead of letters."

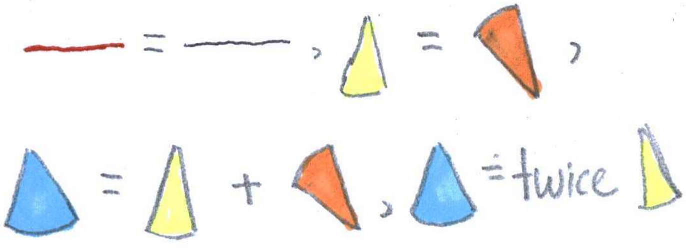
London (1847)

Euclids III.20

The angle at the centre of a circle, is double the angle at the circumference, when they have the fame part of the circumference for their bafe.



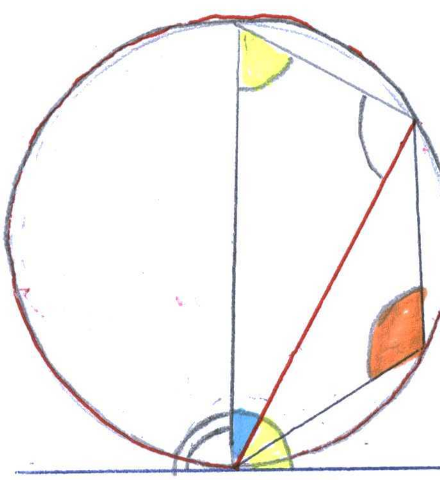
Let the centre of the circle be on a fide of . Becaufe



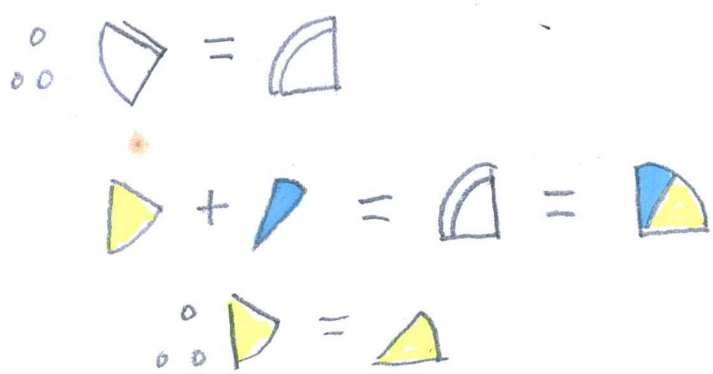
Euclids III.32

If a right line be tangent to a circle, and from the point of contact a right line be drawn cutting the circle, the angle made by this line with the tangent is equal to the angle in the alternate fragment of the circle."

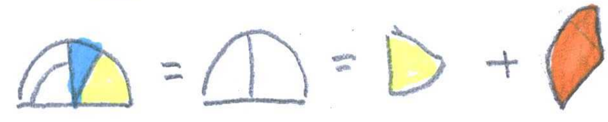
If the chord ſhould paſs through the centre, it is evident the angles are equal, for each of them is a right angle.

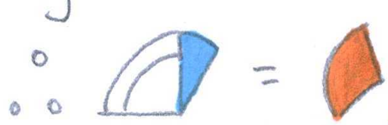


But if not, draw — $\perp$ —— from the point of contact, it muſt paſs through the centre of the circle,



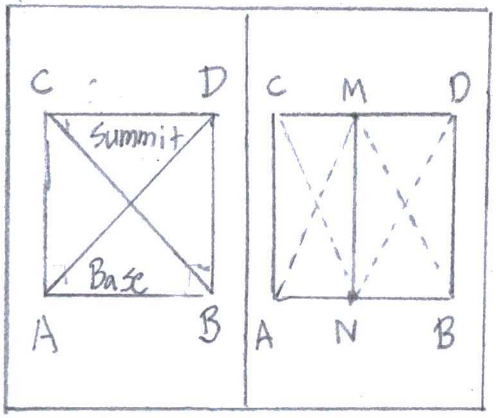
Again





which is the angle in the alternate ſegment.

27.



"Saccheri Quadrilateral"

(from exercise) "Two Saccheri quadrilaterals are defined to be congruent if all their corresponding parts are congruent - their bases, their summits, their sides, and summit angles."

Proof:

Hypothesis	Evidence
1) $\triangle DBA \cong \triangle CAB$	Axiom C-4 (SAS)
2) $\triangle DCB \cong \triangle CDA$	Axiom C-4 (SAS)
3) M and N are the midpoints of CD and AB	Proposition 4.3 (midpoints)
4) $\triangle ACM \cong \triangle BDM$	Axiom C-4
5) $\angle ANM \cong \angle BNM$	(pg 199) "Definition - corresponding angles"
6) $\angle ANM \cong \angle BNM = 90°$	(pg 19) "Definition - supplementary angles"
7) $\triangle ACN \cong \triangle BDN$	Axiom C-4
8) $\triangle CNM \cong \triangle DNM$	Proposition 3.22 (SSS criterion)

10) $\angle CMN \cong \angle DMN = 90°$	(pg 19) "Definition- Supplementary angle"
11) $MN \perp AB$ and $MN \perp CD$	Argument from supplementary.

28. (from exercise) "If Pasch's theorem applies to one pair of opposite sides, then so does the other pair of opposite sides."

Proof:

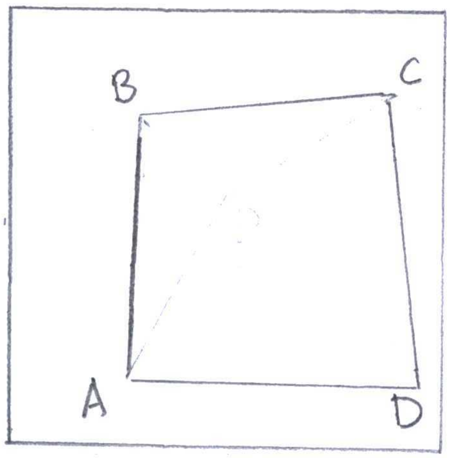
Hypothesis	Evidence
1) A quadrilateral $\square ABCD$ forms from opposite Sides AB and CD, also AC and BD.	(pg 199) "Definition- Quadrilateral"
2) AB and CD are convex	(pg 199) "Definition- Convex"
3) AC and BD are convex	(pg 199) "Definition- Convex"
4) AB and CD match Paschs Theorem	RAA hypothesis

5) AC and BD also follow Pasch's theorem

(from exercise) The following "statements are equivalent: (a) The quadrilateral is convex

(b) Each vertex of the quadrilateral lies in the interior of the opposite angle."

(c) The diagnols of the quadrilateral meet."



"convex quadrilateral"

Proof:

Hypothesis	Evidence
1) A quadrilateral $\square ABCD$ is convex with $\angle B$ the greatest angle	(pg 199) "Definition-Convex"
2) BA, BD, and BC are rays, in addition to AB, AC, and AD	(pg 10) "Definition-rays"
3) $\overrightarrow{BD}$ is between	

two rays, and AC between two rays.	Converse to crossbar theorem.
4) BD intersects AC at a central point	Proposition 2.1
5) D is interior to $\measuredangle B$, C is interior to $\measuredangle A$, B is interior to $\measuredangle D$ and A is interior to $\measuredangle C$.	(pg 115) "Definition - interior"

(from exercise) "Saccheri and Lambert Quadrilaterals are convex."

Proof:

Hypothesis	Evidence
Steps #1-5 above	

29. (from exercise) "The interior of a convex quadrilateral is a convex set and that the point of intersection of the diagnols lies in the interior."

Proof:

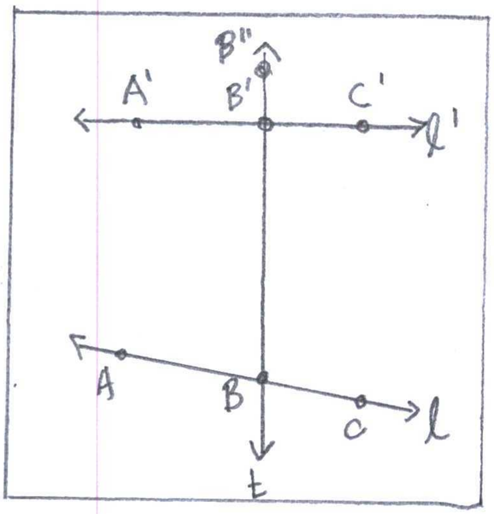
Hypothesis	Evidence

1) A quadrilateral □ABCD is convex with ∡B the greatest angle	(pg 149) "Definition - convex"
2) BA, BD, and BC, in addition to AB, AC, and AD	(pg 19) "Definition - rays"
3) $\overrightarrow{BD}$ is between two rays, and AC between two rays	Converse to crossbar theorem
4) BD intersects AC at an interior point	(pg 115) "Definition - interior"
5) □ABCD has a convex set in the interior from a central point and BD and AC	(pg 149) "Definition - convex"

30. (from exercise) "Saccheri and Lambert Quadrilaterals obey Paschs Theorem"

Proof: Hypothesis	Evidence
1) A quadrilateral □ABCD has an interior point	(Pg 199) Exercise 29
2) A ray from the central point either has: A and D on opposite sides, D and C, B and C, or A and B	Axiom B-4
3) The ray intersects a single segment AD, DC, BC or AB.	Proposition 2.1
4) From an exterior point, then a ray has $\binom{4}{2}$ combinations with points on opposite sides.	Axiom B-4.
5) Saccheri and Lambert quadrilaterals follow Pasch's Theorem.	Pasch's Theorem by argument.

31.



"Corresponding angles"

(from exercise) "Corresponding angles are congruent if and only if alternate interior angles are congruent."

Proof:

Hypothesis	Evidence.
1) Line t cuts through lines l at point B and l' at point B'	Proposition 2.1
2) Points A and C are on opposite sides t, so A*B*C	Axiom B-4
3) Points A' and C' are on opposite sides t, so A'*B'*C'	
4) $\measuredangle AB'B = \measuredangle CBB'$	RAA Conjecture to alternate interiors
5) $\measuredangle AB'B + \measuredangle AB'B'' = 180°$	(pg) "Definition- Supplementary angle"
6) $\measuredangle CBB' + \measuredangle ABB' = 180°$	(pg 19) "Definition- supplementary angle"
7) $\measuredangle AB'B'' = \measuredangle ABB'$	RAA outcome for corresponding angles.

33.

a) Complement of an acute angle - a 90° rotation around the vertex minus the angle.

b) Triangle inequality in a Non-Archimedean plane - any two sides of a triangle are greater than the third."

Please review exercise 14.

Corollary 2 to Exterior Angle Theorem -
"The sum of two angles of a triangle are less than two right angles."

Proof

Hypothesis	Evidence
1) A triangle forms from three points: A, B, C	(pg 98) "Definition - Triangle"
2) On line AB is point D for $A*B*C$	Proposition
3) $(\measuredangle CBD)^\circ > (\measuredangle A^\circ)$	Exterior Angle (EA) Theorem 4.2
4) $(\measuredangle CBD)^\circ + (\measuredangle CBA)^\circ = 180^\circ$ $> (\measuredangle A)^\circ + (\measuredangle CBA^\circ)$	(pg 19) "Definition - supplementary"

c) (from exercise) "The angle sum of every triangle is greater than a 'straight' angle."

The statement above defines an elliptical geometry. A proof is visible by protractor,